

Tutorial Questions (Set-III)

Q1 Determine the derived set of the following subset of the real line.

- (i) $\{\frac{1}{n} : n \in \mathbb{N}\}$.
- (ii) $\{\frac{1}{n} + \frac{1}{m} : n, m \in \mathbb{N}\}$.
- (iii) $\{\frac{a}{2^n} \in \mathbb{Q} : n \in \mathbb{N} \text{ and } 0 \leq a \leq 2^n\}$.
- (iv) Cantor ternary set.

Q2 Let (X, d) be a metric space. Prove the following statements.

- (i) An open ball is an open set and a closed ball is a closed set.
- (ii) An arbitrary union of open sets in (X, d) is open.
- (iii) A finite intersection of open sets in (X, d) is open.
- (iv) A subset $F \subseteq X$ is closed if and only if $X \setminus F$ is open in (X, d) .
- (v) If $\{x_1, x_2, \dots, x_n\} \subseteq X$, then $X \setminus \{x_1, x_2, \dots, x_n\}$ is open.
- (vi) For a metric subspace $(Y, d|_Y)$ of (X, d) , a subset $G \subseteq Y$ is open in $(Y, d|_Y)$ if and only if $G = Y \cap U$ for some open set U in (X, d) . Also, $F \subseteq (Y, d|_Y)$ is closed if and only if $F = Y \cap K$ for some closed set K in (X, d) .

Q3 Let $\mathbb{R}$ be the real line and $a \in \mathbb{R}$. Consider the maps $T_a: \mathbb{R} \rightarrow \mathbb{R}$ and $\mu_a: \mathbb{R} \rightarrow \mathbb{R}$ given by

$$T_a(x) = x + a \quad \text{and} \quad \mu_a(x) = ax, \quad \forall x \in \mathbb{R}.$$

- (i) Show that T_a is a homeomorphism and $(T_a)^{-1} = T_{-a}$.
- (ii) If $a \neq 0$, then show that μ_a is a homeomorphism and $(\mu_a)^{-1} = \mu_{a^{-1}}$.
- (iii) Let $\mathbb{R}^2$ be the Euclidean plane. Consider the maps *plus*: $\mathbb{R}^2 \rightarrow \mathbb{R}$ and *dot*: $\mathbb{R}^2 \rightarrow \mathbb{R}$ given by *plus*(x, y) = $x + y$ and *dot*(x, y) = xy , for $x, y \in \mathbb{R}$. Show that *plus* and *dot* are both continuous maps.

Q4 Let (X, d) be a metric space. Prove the following statements.

- (i) If d is discrete, then X is compact if and only if X is finite.
- (ii) $[0, 1] = \{x \in \mathbb{R} : 0 \leq x \leq 1\}$ is a compact subset of the real line $\mathbb{R}$.
- (iii) A compact subset of (X, d) is closed and bounded.
- (iv) A sequentially compact subset of (X, d) is compact.
- (v) A closed and bounded subset of the Euclidean space $\mathbb{R}^n$ is compact. (This is known as *Heine-Borel theorem*)

- (vi) Without using Heine-Borel theorem, deduce that the subsets $\{0, \frac{1}{n} : n \in \mathbb{N}\}$ and $\{0, \frac{1}{n}, \frac{1}{n} + \frac{1}{m} : n, m \in \mathbb{N}\}$ are compact subsets of $\mathbb{R}$.

- Q5 (i) Show that $[0, 1]$ is a connected subset of the real line $\mathbb{R}$.
(ii) Show that a subset I of $\mathbb{R}$ is connected if and only if I is an interval.
(iii) Show that a metric space (X, d) is disconnected if and only if there is a surjective continuous map $\phi: X \rightarrow \{0, 1\}$, where $\{0, 1\}$ is the discrete metric space.
(iv) Show that a path-connected metric space is connected. Further, show that $\mathbb{R}^n$ and convex subsets of $\mathbb{R}^n$ are path-connected.

Q6 Let (X, d) be a metric space and $\emptyset \neq A \subseteq X$. Let $d_A: X \rightarrow \mathbb{R}$ be a function given by

$$d_A(x) = \text{dist}(x, A) = \inf\{d(x, a) : a \in A\}, \quad \forall x \in X.$$

- (i) Show that d_A is uniformly continuous.
(ii) Let $\emptyset \neq S$ be closed and $x_0 \in X \setminus S$. Show that there is a continuous function $f: X \rightarrow \mathbb{R}$ such that $f(x_0) = 1$ and $f(S) = \{0\}$.
(iii) Let S and T be non-empty disjoint subsets of X . Show that there is a continuous function $g: X \rightarrow \mathbb{R}$ such that $g(S) = \{0\}$ and $g(T) = \{1\}$. Further, we may choose $g(X) \subseteq [0, 1]$.
Q7 (i) Let $f: (X, d) \rightarrow (Y, \rho)$ and $g: (Y, \rho) \rightarrow (Z, \sigma)$ be continuous functions. Show that the composition $g \circ f: (X, d) \rightarrow (Z, \sigma)$ is continuous.
(ii) Let $f: (X, d) \rightarrow \mathbb{R}$ and $g: (X, d) \rightarrow \mathbb{R}$ be real-valued continuous functions on the metric space (X, d) . Show that $f + g$ and fg given by

$$(f + g)(x) = f(x) + g(x) \quad \text{and} \quad (fg)(x) = f(x)g(x), \quad \forall x \in X$$

are continuous. Also, if $g(x) \neq 0$ for all $x \in X$, then $\frac{f}{g}$ given by

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}, \quad \forall x \in X$$

is continuous.

Q8 (Optional)

- (i) Show that a perfect, complete metric space is uncountable.
(ii) Show that the Cantor ternary set is perfect. Also, construct a bijection between the Cantor ternary set and $[0, 1]$.